

CS Life Lesson: If your question has a graph, great!
If it doesn't, make it have a graph :)

Types of Graphs:

Simple undirected

By default in this class

Directed

Multigraph

Definition (Simple Undirected Graph): A tuple (V, E) where
 $\hookrightarrow V$ is a finite set of vertices
 $\hookrightarrow E$ is a set called the set of edges s.t.
 every element of E is a $\{u, v\}$ for distinct u, v .

Example: $V = \{v_1, v_2, v_3, v_4, v_5, v_6\}$ $E = \{\{v_1, v_2\}, \{v_1, v_4\}, \{v_2, v_3\}, \{v_3, v_4\}, \{v_5, v_6\}\}$

Picture:

Adjacency matrix:

	v_1	v_2	v_3
v_1	0	1	0
v_2	1	0	1
v_3	0	1	0

Notation: $n =$ num vertices in graph $|V|$ $m =$ number of edges $|E|$

Suppose $e = \{u, v\}$ is an edge. We say:

- $\hookrightarrow u$ and v are endpoints of e
- $\hookrightarrow u$ and v are adjacent
- $\hookrightarrow u$ and v are incident on e
- $\hookrightarrow u$ is a neighbour of v , v is a neighbour of u

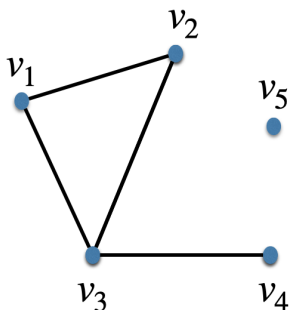
Definition (neighborhood):

$$N(v) = \{u \in V : \{v, u\} \in E\}$$

Edge cases:

$E = \emptyset$ possible

$V = \emptyset$ not possible, a graph must have at least one vertex.



$$N(v_1) = \{v_2, v_3\}$$

$$N(v_3) = \{v_1, v_2, v_4\}$$

$$N(v_5) = \emptyset$$

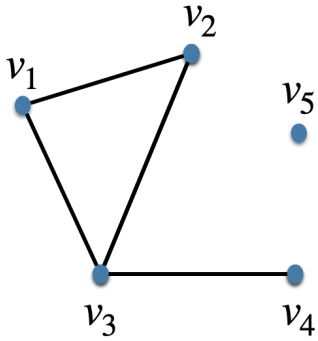
Definition (degree): size of neighbourhood.

Definition (d -regular graph): every vertex has exactly d degrees
 $(\deg(v) = d) \forall v \in V$

Lemma (Handshake Lemma): Let $G = (V, E)$ be a graph. Then,

Proof:

$$\sum_{v \in V} \deg(v) = 2m$$



Place tokens on edges. Each vertex puts a token on each edge.

Each edge gets 2 tokens.

We will now double count num tokens.

$$1. \sum_{v \in V} \deg(v)$$

$$2. 2m$$

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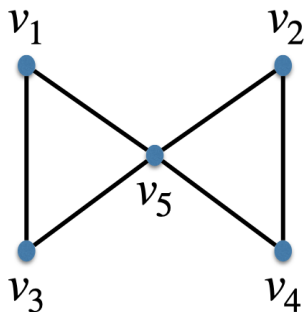
Question: Is it possible to have a graph with 251 vertices in which each vertex is adjacent to exactly 5 other vertices? No. 251×5 is not divisible by 2

Definition (walk): A sequence of vertices $u_0, u_1, u_2, \dots, u_k$ $k \geq 0$ s.t. $\{u_{i-1}, u_i\} \in E \quad \forall i \in \{1, 2, \dots, k\}$

Definition (path): A walk with no repeat vertices.

Fact: $\exists$ a path from v to $v' \iff \exists$ a walk from v to v'

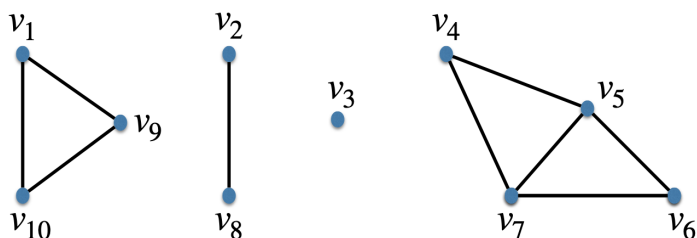
Definition (circuit): A walk that starts and ends at the same vertex



Definition (cycle): A circuit with no repeated vertices except start and end (length ≥ 3)

Definition (acyclic graph): A graph with no cycles

Definition (connected graph): $\exists$ a path from any two vertices in the graph.



This graph has 4 **connected components**:

A graph is connected if and only if: $\exists$ a path between any 2 vertices.

Question: What is the least number of edges needed to connect n vertices?

Theorem: min edges needed $m \geq n-1$, and $m = n-1 \Leftrightarrow G$ is acyclic

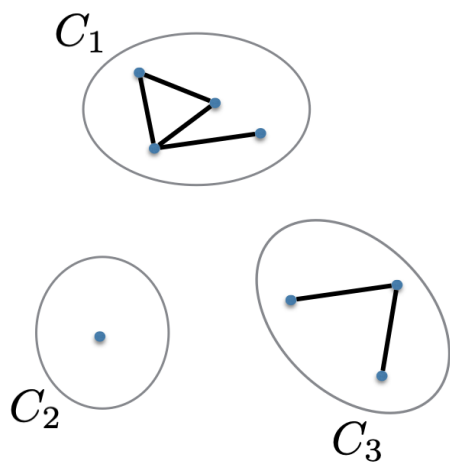
Proof: Imagine the following process:

- Remove all edges from G .
- add them back one by one in some arbitrary order.

n isolated vertices $\rightarrow G$

n connected components (CCs) $\rightarrow 1$ CC

Consider a step of adding an edge back.



2 possibilities:

↳ connector edge — connects 2 CCs.
 ↳ #CCs goes down by 1.
 ↳ cannot create new cycle.

↳ cycle-creator edge — an edge within a CC.
 ↳ doesn't change #CCs
 ↳ creates a cycle 🍷

n CCs $\xrightarrow{i}$ $n-1$ CCs $\xrightarrow{i}$ $n-2$ CCs $\rightarrow \dots \rightarrow 1$ CC
 must have at least $n-1$ type (i) edges.

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Definition (tree): Acyclic, connected graph where $m = 2n - 1$.
Satisfies 2 of the 3 conditions stated.

leaf:

internal node:

rooted tree:

Minimum Spanning Tree (MST) Algorithm

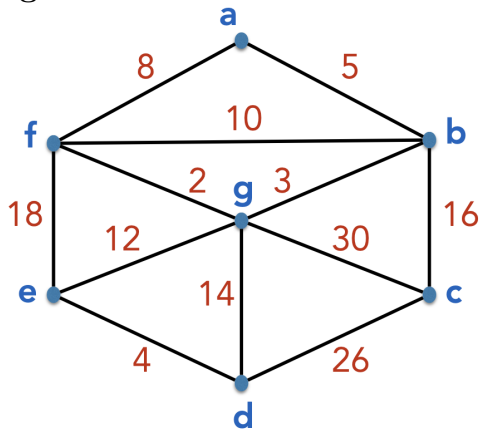
Input:

Output:

Observation:

Convenient assumption:

Algorithm:



Lemma (MST Cut Property): Let $G = (V, E)$ be a connected graph with distinct positive edge weights.

Proof Idea: Proof by contradiction.

Let T be the MST.

